CAT

QUESTION ONE (a)

Concept of Search in AI

Search is the process of finding a solution, exploring possible paths and moving from an initial state to a goal state.

AI systems use search to: solve puzzles, navigate maps, play games, find shortest paths and make decisions. Suppose Google Maps wants to find the shortest route from Nairobi CBD → MMU, the AI checks possible roads, compares distances and chooses best route.

Uninformed Search vs Informed Search.

Uninformed Search (Blind Search): These algorithms have no additional information about states beyond that provided in the problem definition. They explore the search space systematically e.g., Breadth-First Search and Depth-Search-First.

Informed Search (Heuristic Search): These use problem-specific knowledge that is, heuristics, to find solutions more efficiently. They guess which path is closest to the goal. Eg. A* Search and Greedy Best First Search.

Differences between the two:

Feature	Uninformed Search	Informed Search
Knowledge	No knowledge of the goal distance.	Uses heuristics to estimate distance.
Efficiency	Slower; explores many nodes.	Faster; more focused.
Complexity	Generally higher time/space complexity.	Lower complexity if heuristic is good.
Examples	BFS, DFS, Uniform Cost Search.	A*Search, Greedy Best-First Search.

QUESTION ONE (b).

A* is the gold standard for pathfinding because it uses both the actual cost ($g(n)$) and an estimated cost to the goal (h(n)). The formula is: f(n) = g(n) + h(n).Find the program in question1-b.py file.

QUESTION ONE(c).

(i) Distinguish

1. Valid: A formula is always true under *all* possible interpretations -a Tautology.

2. **Satisfiable:** A formula is true under *at least one* interpretation.

3.**Unsatisfiable:** A formula is *always false under* all interpretations. - a Contradiction.

4.**Falsifiable:** A formula is false under *at least one* interpretation.

(ii)Truth Table.

P	**Q**	**R**	**P∧Q**	**¬R**	**P∧Q→¬R**
T	T	T	T	F	**F**
T	T	F	T	T	**T**
T	F	T	F	F	**T**
T	F	F	F	T	**T**
F	T	T	F	F	**T**
F	T	F	F	T	**T**
F	F	T	F	F	**T**
F	F	F	F	T	**T**

Truth table analysis:

The formula is **NOT Valid** –because it is not always true.

The formula is **Satisfiable** –because it is true in some rows.

The formula is **NOT unsatifisfiable** – Because it is not always false .

The formula is **Falsifiable** –Because false exists in the first row.

QUESTION TWO(a).

PEAS and PAGE for DNA Sequencing System

For an agent identifying Shakahola victims, we define its environment and properties:

PEAS Framework:

P-**Performance Measure:** Accuracy of DNA matching (%), speed of identification, minimization of false positives.

E-**Environment:** Digital database of corpse DNA, relative DNA samples, Ministry of Health servers, crime scene.

A-**Actuators:** Display screen for matches, automated report generator, notification system to officials.

S-**Sensors:** DNA sequence scanners, file upload interface, database query results.

PAGE Framework:

P-**Percepts:** Raw genomic sequences, metadata (location of sample, age).

A-**Actions:** Compare sequences, flag discrepancies, output "Match Found" or "No Match".

G-**Goals:** Correctly identify all victims and link them to respective families.

E-**Environment:** Software-based, highly secure, data-intensive.

QUESTION TWO (b)

Vacuum Cleaner Agent

Find the implementation of vacuum cleaner agent in vacuum_agent.py file.